

iEat: Dietary monitoring with bio-impedance sensing

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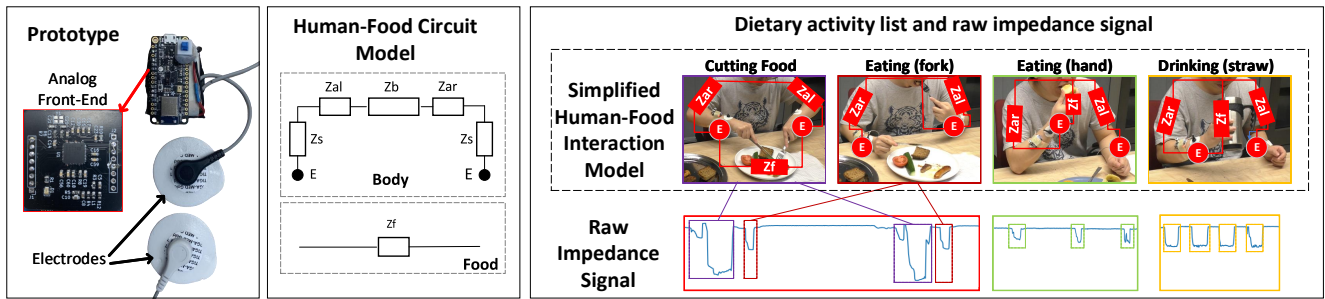


Figure 1: The hardware prototype and raw signal of the dietary activity. (in the simplified human-food circuit model: Zar and Zal are the impedance of the right arm and left arm, Zs are the surface impedance on the skin)

ABSTRACT

We explore an atypical use of bio-impedance by leveraging the unique temporal impedance patterns caused by the dynamic circuit changes between a pair of electrodes due to the body motions, and interactions with metal utensils and food during dining activities. Specifically, we present iEat, a wearable impedance-sensing device for automatic food intake monitoring without using external devices such as smart utensils. Using only one impedance channel with one electrode on each wrist, iEat detects food intake activities (e.g. cutting, putting food in the mouth with or without utensils, drinking, etc.) and food types from a defined category. At idle, iEat measures the normal body impedance between the wrists; while eating, new parallel circuits will be formed between the hands through the utensils and food. To quantitatively evaluate iEat in real-world settings, a food intake experiment was conducted including 40 meals performed by ten volunteers in a realistic table-dining environment. With a light-weight convolutional neural network and leaving one subject out cross-validation, iEat could detect five food intake-related activities with 86.27 % average accuracy, and classify eight types of foods with 77.73 % average accuracy.

CCS CONCEPTS

• Human-centered computing → Ubiquitous computing.

KEYWORDS

Bio-impedance Sensing, Automatic Dietary Monitoring, Human-food Interaction

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